Part 13 — The gaps: separations, deconstruction, enzymology, consortia, measurement and control

Six supplementary chapters covering what an audit of the tracker found missing. They are numbered S1 to S6 rather than by week, because each extends a week that already exists rather than replacing it, and the heading of each says which. The order is mine and differs from the audit's: I put downstream separations first because it is the largest and most defensible gap, and formal control theory last because it serves argument more than it serves the work.

The audit was broadly right and partly wrong. Right about separations, deconstruction, two-substrate kinetics, consortia, untargeted and absolute-quantitative omics, optimal control and genetic code expansion. Wrong that biosafety and dual use are absent — part 9 chapter 32 covers containment, the institutional committee and the Korean LMO Act, and part 5 chapter 53 works the opioid case — and imprecise about bistability, which part 8 chapter 26 and part 7 chapter 24 treat informally; what was actually missing there is the eigenvalue analysis, which chapter S6 supplies. Two items I have left out deliberately: genetic code expansion and quantitative genetics, which are real gaps but are reading rather than calculation, and the Tier 3 list, which is a literature-awareness exercise.

Everything quantitative here was computed, with the parameters printed beside it, and the assumptions that carry the most weight are named at the end of each chapter. Two numbers in chapter S1 and one in S2 are deliberately presented as sensitivity ranges rather than values, because the inputs — resin price and cellulase price per kilogram of protein — move by an order of magnitude between sources and a single figure from me would be worse than a range.

Ch.	Extends	Chapter
S1	wk 44, 49	Downstream processing: choosing and sizing a train
S2	wk 55–57	Biomass deconstruction: pretreatment, saccharification and the severity trade
S3	wk 3	Two-substrate kinetics and allostery: mechanisms, MWC and KNF
S4	wk 9, 26–27	Synthetic consortia: division of labour, and why it does not stay divided
S5	wk 43, 51	Measuring the pools and the proteins: quantitative metabolomics and proteomics
S6	wk 16, 27, 44	Control theory for pathways: stability, bistability, integral feedback, optimal feeding

Chapter S1 · extends weeks 44 and 49

Downstream processing: choosing and sizing a train

The module has told you repeatedly that separation decides whether a titer is worth having. It has never told you how to choose the train or size a column. This chapter does, and the first result is that recovery is a product.

Recovery is multiplicative, and that is the whole design constraint

Every unit operation loses product. Overall recovery is the product of step recoveries, so the number of steps matters more than the quality of any one:

Recovery per step	3 steps	5 steps	8 steps	12 steps
95%	86%	77%	66%	54%
90%	73%	59%	43%	28%
80%	51%	33%	17%	7%

Read it the other way and it becomes a specification: to deliver 90% overall you need 96.5% per step in a three-step train, 97.9% in five and 98.7% in eight. **Shortening the train is almost always worth more than improving a step**, which is the opposite of where effort usually goes, and it is the reason an upstream decision that removes a step — secretion instead of retention (part 6 chapter 59), a product that crystallises, in situ removal that delivers a partially purified stream (part 3 chapter 46) — is worth more than any chromatography optimisation.

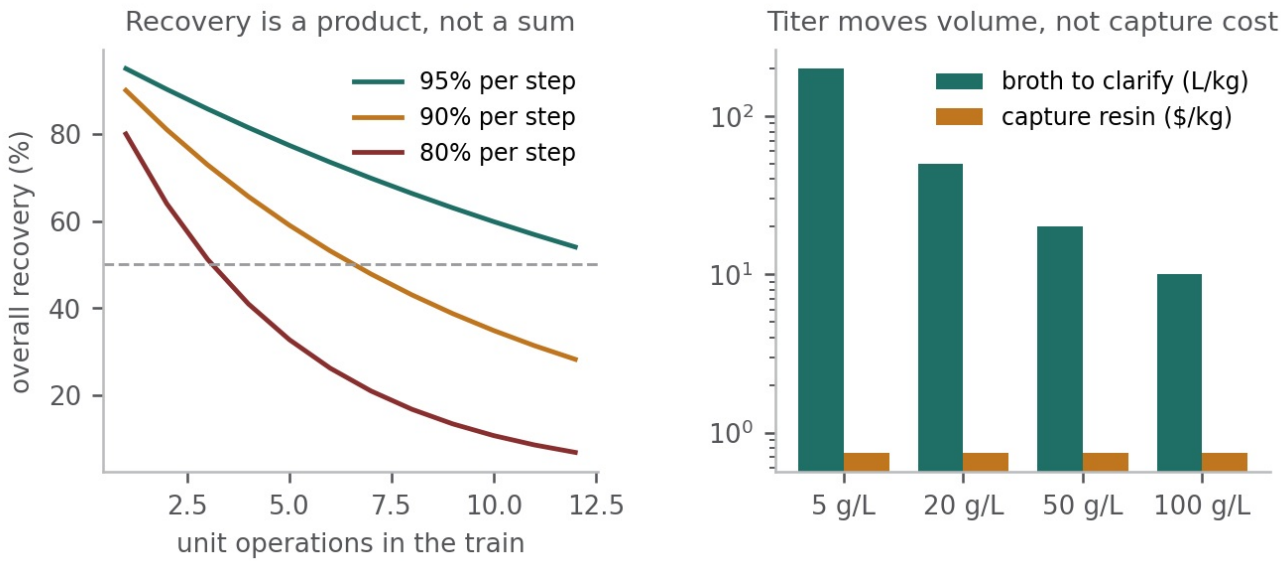


Figure S1.1 Left: overall recovery against train length for three per-step recoveries. Right: what titer does and does not move — the broth volume to clarify scales as 1/titer while the capture resin cost per kilogram does not, for the reason derived below. Computed for this module.

Choosing the train from the product's properties

The sequence is not a matter of taste. Each operation exploits one physical property, so the product's properties select the operations and their order: remove the largest contaminant first, work from high-volume cheap steps to low-volume expensive ones, and never put a capacity-limited step before a volume-reduction step.

Property of the product	What it selects	What it rules out
Secreted, soluble	Clarify, then capture directly from broth	No lysis, no debris removal — two steps saved
Intracellular, soluble	Harvest, lyse, clarify, then capture	Adds the hardest solid–liquid separation in the sequence
Intracellular, insoluble (inclusion body, polymer granule)	Harvest, lyse, wash the solids; for PHA, digest the non-polymer biomass	Chromatography is irrelevant — this is a solids process
Charged at process pH (organic acid, amino acid)	Ion exchange, electrodialysis, reactive extraction, crystallisation of a salt	Solvent extraction of the free species (part 3 chapter 46)

Volatile	Distillation, stripping, pervaporation	Anything requiring the product to stay in solution
Hydrophobic, uncharged	Liquid–liquid extraction, adsorption onto polymer	Ion exchange
Sparingly soluble at some pH or temperature	Crystallisation — the cheapest purification there is	—
Heat-labile	Membranes, chromatography, lyophilisation	Distillation, spray drying at high inlet temperature
A protein needing >99% purity	Capture, intermediate purification, polish — three chromatographic steps	Single-step purification; the purity is not reachable

Sizing the first solid–liquid separation

Cells are small and nearly neutrally buoyant, which is the whole difficulty. Stokes' law for a 1 µm cell with a 100 kg/m³ density difference in water gives a settling velocity of **5.5×10^{-8} m/s — 0.2 mm per hour**. A centrifuge is rated by its sigma factor, the equivalent gravitational settling area that would do the same duty: $\Sigma = Q/v_g$. For 1 m³/h you need **$\Sigma \approx$ 5,100 m²**, and for 10 m³/h, 51,000 m².

A disc-stack centrifuge reaches sigma factors of this order by stacking many short settling paths at high g, which is why it exists. The arithmetic also explains two things the literature mentions and rarely derives: velocity goes as the *square* of diameter, so a 0.5 µm cell needs four times the sigma, which is why some hosts are far harder to harvest than others; and flocculation works by raising the effective diameter, so it buys a quadratic improvement for a small reagent cost. Microfiltration is the alternative, where the duty is set by permeate flux against transmembrane pressure and the enemy is the gel layer, so the design variable is crossflow velocity rather than area.

Sizing a chromatography step, and the cost result that surprised me

A bind-elute step is sized by **dynamic binding capacity** — grams of product bound per litre of resin at the chosen residence time and at a stated breakthrough, conventionally 10%. It is lower than the static capacity, often by half, because mass transfer into the bead limits it; the tool is a breakthrough curve, and the usable figure comes from it rather than from the brochure.

For a 10 m³ batch:

Titer	Product in the batch	Resin at DBC 40 g/L	At DBC 60 g/L	Cycles on a 100 L column at DBC 50
5 g/L	50 kg	1,250 L	833 L	10
20 g/L	200 kg	5,000 L	3,333 L	40
50 g/L	500 kg	12,500 L	8,333 L	100

Either you buy the resin or you cycle a smaller column, and cycling is what everyone does — which turns column volume into campaign time and buffer volume. At ten column volumes of buffer per cycle, the 50 g/L case consumes 100 m³ of buffer, which is itself a utility, a cold room and a waste stream.

Now the result I did not expect until I wrote it down. The resin needed scales with the product mass, and the cost is amortised over the resin's cycle life, so

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capture cost per kg of product = resin price per litre / (DBC in g/L × cycles the resin survives)
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and **the titer cancels**. Four cases, as a sensitivity because resin prices span an order of magnitude:

Resin class	Price	DBC	Cycle life	Capture cost per kg
Ion exchange, cheap	$300/L	50 g/L	200	$0.03
Ion exchange, dearer	$1,000/L	50 g/L	200	$0.10
Affinity-like	$3,000/L	40 g/L	100	$0.75
Protein-A class	$8,000/L	40 g/L	200	$1.00

So raising titer does not reduce the capture cost per kilogram at all; what it reduces is the broth volume to clarify, the buffer, the tank farm and the campaign time. The levers on the capture step itself are **cycle life** and **DBC** — each doubling halves the cost — and both are measured, not quoted. Part 4 chapter 49 modelled separation as a cost scaling with 1/titer, which is right for the volume-driven steps and wrong for capture; treat that chapter's separation line as the volumetric part only.

Crystallisation, drying, and the cheap end

Crystallisation is the cheapest purification available and the most underused in fermentation papers: it combines purification and concentration in one step, and its yield follows directly from the solubility curve — cool or acidify from a concentration C_1 to a solubility C_2 and the yield is $(C_1 - C_2)/C_1$. Supersaturation has to be controlled, because the metastable zone decides whether you get crystals you can filter or fines you cannot; seeding at a controlled supersaturation is how that is done. For an organic acid, acidifying below the pK_a to crystallise the free acid is the standard route and is the reason part 3 chapter 45's pH arithmetic matters downstream as well as upstream.

Drying closes the train, and the comparison is thermal: spray drying is cheap and fast and exposes the product to a hot inlet; lyophilisation is gentle and costs roughly an order of magnitude more energy per kilogram of water removed. Which you need is set by heat lability, and it should be decided before the formulation, not after.

Recovery	`overall = Π(step recoveries); 90% over 8 steps needs 98.7% each`
Centrifuge duty	$\Sigma = Q/v_g$; $v_g = \Delta\rho \cdot g \cdot d^2/18\mu = 5.5 \times 10^{-8}$ m/s for a 1 µm cell
Chromatography sizing	`resin volume = mass/DBC; cycles = mass/(DBC × column volume)`
Capture cost	`resin price ÷ (DBC × cycle life) — independent of titer`
Crystallisation yield	$(C_1 - C_2)/C_1$ from the solubility curve
What titer moves	`volume-driven steps only: clarification, buffer, tankage, campaign time`

Your product is a secreted 25 kDa protein at 8 g/L, needed at 98% purity, and heat-labile. Design the train and name its largest cost.

Four steps, and the aim is to keep it to four. **Clarify** by disc-stack centrifugation followed by depth filtration — 125 litres of broth per kilogram of product at this titer, so the duty is set by volume and the sigma arithmetic above. **Capture** by ion exchange, chosen over affinity because the cost table says affinity costs ten to thirty times more per kilogram and 98% does not need it; size it from a measured dynamic binding capacity at your residence time, not the brochure's static figure. **Polish** by a second chromatographic step on an orthogonal property — hydrophobic interaction after ion exchange — since 98% from a single capture step is not reachable. **Formulate** by ultrafiltration/diafiltration into the final buffer and lyophilise, because spray drying is ruled out by heat lability.

Overall recovery at a realistic 90% per step is 66%, which is the number to carry into the techno-economic model — and the reason to ask whether the polish step can be avoided by a better capture. The largest cost is not the resin: at $1,000/L, DBC 50 and 200 cycles it is $0.10/kg. It is the **buffer and the water** — tens of cubic metres per batch, with the cold storage, the WFI-grade preparation if this is a biologic, and the waste — plus the lyophiliser, which is capital-intensive and slow and therefore sets the plant's throughput. That inversion, where the exotic-sounding step is cheap and the utilities dominate, is the usual finding and the reason to cost a train rather than intuit it.

Three things people get wrong

- ‘We will improve the chromatography yield.’ Removing a step is worth more: 90% per step over eight steps is 43% overall.
- ‘Higher titer makes purification cheaper per kilogram.’ It makes the volume-driven steps cheaper. Capture cost per kilogram is resin price ÷ (DBC × cycle life) and does not contain titer.
- ‘Chromatography is how you purify.’ For an intracellular polymer or a crystallisable acid it is irrelevant, and crystallisation purifies and concentrates in one step.

Self-check — answer on paper first

1. A train of six steps recovers 55% overall. Where do you look first?

At the number of steps, then at the worst step. Six steps at 55% implies about 90.5% each if they are equal, which is unremarkable — so the question is whether any step can be deleted, combined with its neighbour, or moved upstream where the volume is larger and the loss cheaper. Only then look for the one step well below the others, which a step-by-step mass balance identifies; and note that a step losing product to a side stream may be recoverable by recycle, which is a different fix from improving the step itself.

2. Derive why the capture cost per kilogram does not depend on titer, and say what that implies for a project plan.

The resin volume needed is the product mass divided by the dynamic binding capacity, so it scales with mass; the cost is the resin price times that volume, amortised over the number of cycles the resin survives; dividing by the mass processed cancels the mass entirely, leaving price ÷ (DBC × cycle life). It implies that a titer campaign should be

justified by volume-driven costs — clarification, buffer, tankage and campaign time — and that the two things worth measuring early for the capture step are the dynamic binding capacity at your real residence time and the resin's lifetime in your real feed, since fouling is what actually ends a resin's life.

3. Why can a 0.5 µm host be much more expensive to harvest than a 1 µm one?

Because settling velocity goes as the square of the diameter, so halving the diameter quarters the velocity and quadruples the required sigma factor — a bigger or slower centrifuge, or a switch to microfiltration where the flux is set by fouling. It is also why flocculation is attractive: raising the effective particle diameter by aggregation buys a quadratic improvement for the cost of a polymer, at the price of a contaminant in the product stream that must then be removed.

What to open a book for. Harrison, Todd, Rudge & Petrides, *Bioseparations Science and Engineering*, which is the standard text and is organised exactly as this chapter is — by property and by step. Doran's chapters on unit operations for the transport side. For chromatography scale-up specifically, a vendor's process-development handbook is more useful than any textbook, and the breakthrough-curve protocol in it is the one measurement this chapter rests on.

Chapter S2 · extends weeks 55 to 57

Biomass deconstruction: pretreatment, saccharification and the severity trade

Part 5 told you which inhibitors to measure in a hydrolysate. It never said where they come from: they are made by the same step that releases the sugar, which makes pretreatment a trade rather than a maximisation.

What is in the biomass, and the sugar ceiling

Lignocellulose is three polymers and some ash. For a representative composition — 40% cellulose, 25% hemicellulose, 22% lignin, 13% ash and extractives — the sugar ceiling follows from hydrolysis adding a water molecule per bond: glucan to glucose multiplies by 180.16/162.14 = 1.111, and xylan to xylose by 150.13/132.12 = 1.136.

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per kg of dry biomass:  0.444 kg glucose + 0.284 kg xylose = 0.729 kg of fermentable sugar
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That ceiling matters because it is the number every downstream yield multiplies. At 0.729 kg of sugar and ethanol's 0.511 g/g from part 2 chapter 18, the biomass-to-ethanol ceiling is 0.373 kg/kg — and only if the pentoses are fermented, which is the catabolite-repression problem of part 1 chapter 9. Realistic saccharification and recovery of 80% brings it to 0.583 kg of sugar and 0.298 kg of ethanol; at 65%, to 0.474 and 0.242.

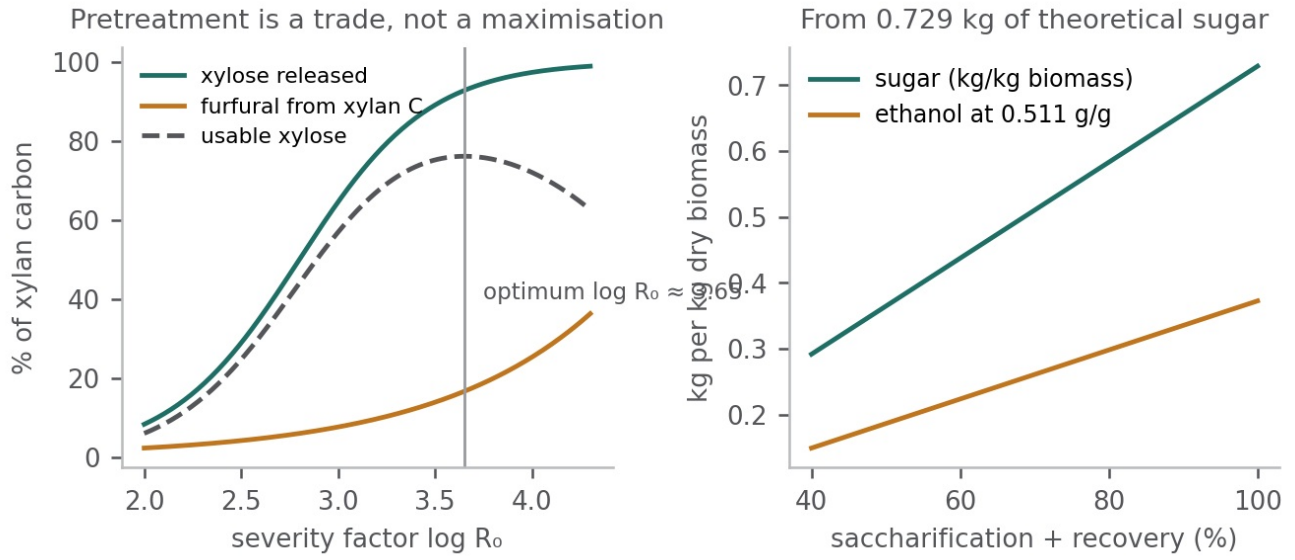


Figure S2.1 Left: the severity trade — xylose released and furfural formed against the severity factor, with the usable difference peaking at log R₀ ≈ 3.5. The two curves are illustrative shapes fitted to the qualitative behaviour, not measured data, and the optimum's position depends entirely on them. Right: sugar and ethanol per kg of dry biomass against the fraction of the 0.729 kg ceiling actually recovered. Computed for this module.

Pretreatment, and the severity factor

Native cellulose is crystalline and sheathed in hemicellulose and lignin, so enzymes cannot reach it: pretreatment exists to raise accessibility. The main routes and what each does:

Pretreatment	What it removes or changes	Cost
Steam explosion	Solubilises hemicellulose, disrupts structure by explosive decompression	Furfural and HMF formation; acetic acid released from hemicellulose
Dilute acid	Hydrolyses hemicellulose to monomeric pentoses	Most degradation products of any route; neutralisation produces gypsum
Alkaline (lime, NaOH)	Removes lignin, swells cellulose	Alkali recovery; lower sugar degradation, so fewer furans
Organosolv	Dissolves lignin into an organic phase, giving a recoverable lignin stream	Solvent recovery and the capital it needs
AFEX (ammonia fibre expansion)	Decrystallises cellulose, leaves the solids nearly dry, few inhibitors	Ammonia recycle

The common currency is the **severity factor**, which collapses temperature and time into one number: log R₀ = log₁₀[t ·

exp((T − 100)/14.75)], with t in minutes and T in °C. It is an Arrhenius-type combination, so a 14.75 °C rise is worth a factor of e in time:

Condition	log R_0	Condition	log R_0
160 °C, 10 min	2.77	190 °C, 2 min	2.95
180 °C, 10 min	3.36	200 °C, 5 min	3.64
210 °C, 3 min	3.72	220 °C, 1 min	3.53

The trade is the chapter's point. Raising severity releases more hemicellulosic sugar and makes cellulose more accessible, and it also degrades the sugar it released: pentoses dehydrate to furfural, hexoses to 5-hydroxymethylfurfural, and the acetyl groups of hemicellulose leave as acetic acid — the four inhibitors part 5 chapter 56 told you to measure. So **the optimum is interior**, and it is set by the ratio of release kinetics to degradation kinetics for your feedstock, not by a number from a paper on a different one. Detoxification — overliming, activated carbon, resins, laccase treatment, or simply evolving a tolerant strain — is the alternative to backing off the severity, and it costs a step in the sense of chapter S1.

Enzymatic saccharification

The cocktail is three activities working in series on an insoluble substrate: **endoglucanases** cut internally and create new chain ends, **cellobiohydrolases** processively release cellobiose from those ends, and **β-glucosidase** hydrolyses cellobiose to glucose — which matters disproportionately because cellobiose inhibits the cellobiohydrolases, so the whole cascade stalls behind a missing β-glucosidase. **Lytic polysaccharide monooxygenases** oxidatively cleave crystalline regions and need a reductant and oxygen, which is why they were missed for decades and why they raised cocktail performance when added. Hemicellulases and accessory esterases remove the sheath.

Three features make the kinetics unlike anything in part 2 chapter 3. The substrate is **insoluble**, so the relevant concentration is accessible surface rather than molarity, and conversion slows as the easy fraction is consumed — the characteristic hyperbolic-with-a-long-tail curve. Enzymes **adsorb** to the substrate, including unproductively to lignin, so part of your dose is lost to a polymer that is not the target. And product inhibition is strong, which is the argument for simultaneous saccharification and fermentation: the yeast consumes glucose as it appears and relieves the inhibition, at the cost of a temperature compromise between the enzymes' optimum near 50 °C and the organism's near 35.

Enzyme cost is the line that decides a cellulosic process, and it is also the number I cannot give you, because the price per kilogram of enzyme protein spans an order of magnitude between sources. As a sensitivity, with 80% saccharification of the composition above:

Loading	$10/kg protein	$50/kg	$200/kg	$500/kg
10 mg/g glucan	$0.13/kg ethanol	$0.67	$2.68	$6.71
20 mg/g glucan	$0.27	$1.34	$5.37	$13.42

Against an ethanol price near $0.60/kg, the table says the process is viable only in its top-left corner — cheap enzyme at low loading — which is exactly why enzyme cost reduction and on-site enzyme production have been the central engineering effort in cellulosic ethanol, and why **consolidated bioprocessing** — one organism that secretes the cocktail and ferments the sugar — is the standing ambition. Its difficulty is the combination: a good cellulase secretor is usually a poor fermenter and vice versa, so CBP is either a cellulolytic organism engineered to ferment or a fermenter engineered to secrete a cocktail, and the second runs into the secretion-capacity limits of part 6 chapter 59.

Sugar ceiling	`1.111 × glucan + 1.136 × xylan; 0.729 kg/kg for 40/25 composition`
Ethanol ceiling	`0.729 × 0.511 = 0.373 kg/kg of dry biomass, pentoses included`
Severity	`log R₀ = log₁₀[t·exp((T−100)/14.75)]; 14.75 °C ≈ a factor of e in time`
The trade	`release and degradation rise together — the optimum is interior`
Cocktail	`endoglucanase + cellobiohydrolase + β-glucosidase + LPMO; cellobiose inhibits`
Enzyme cost	`loading × glucan fraction × price; viable only at low loading and cheap enzyme`

A collaborator proposes raising pretreatment severity from log R_0 3.0 to 3.7 to get more sugar. What do you say, and what would you measure?

That it will indeed release more sugar and that the question is what the extra sugar costs. On the illustrative curves in figure S2.1 the released xylose rises from about 69% to 94% of the xylan while furfural rises from 8% to 18% of the xylan carbon, so the *usable* xylose barely improves and the fermentation inherits more than double the inhibitor load. That load does not appear in the saccharification yield — it appears later as a longer lag, a lower growth rate and a lower final titer, which is where it will be misdiagnosed as a strain problem.

What to measure, in this order: the four inhibitors by HPLC and total phenolics, at both severities, reported as concentrations in the hydrolysate as fed; growth rate and final titer of your actual strain on both hydrolysates, against a synthetic-medium control, which is the only test that matters; and the sugar yield itself, so the trade is quantified rather than assumed. Then the decision is between backing off the severity, detoxifying — which adds a step, with chapter S1's multiplicative penalty — or evolving tolerance, which part 8 chapter 28 says is the method with the better record for a polygenic phenotype like this.

Three things people get wrong

- ‘Maximise sugar release.’ The same step that releases sugar degrades it; the optimum is interior and feedstock-specific.
- ‘The hydrolysate is a sugar solution.’ It is a sugar solution plus furfural, HMF, acetate and phenolics, and the last of those is the least characterised and the most variable.
- ‘Enzyme cost is a detail.’ At a plausible price and loading it exceeds the value of the ethanol it makes.

Self-check — answer on paper first

1. Compute the sugar and ethanol ceiling for a feedstock that is 45% cellulose and 20% hemicellulose.

Glucose 0.45 × 1.111 = 0.500 kg/kg; xylose 0.20 × 1.136 = 0.227; total 0.727 kg of sugar per kg of dry biomass — almost identical to the 40/25 case, because the two polymers convert at nearly the same mass ratio. Ethanol at 0.511 g/g is 0.372 kg/kg if the pentose is fermented, and 0.256 kg/kg if only the glucose is, which is the single largest reason pentose utilisation is a research priority.

2. Why does a missing β-glucosidase stall the whole cocktail rather than just slowing the last step?

Because cellobiose, the product of the cellobiohydrolases, inhibits them. Without β-glucosidase to hydrolyse it to glucose, cellobiose accumulates and shuts down the processive enzymes that are doing most of the work on crystalline cellulose, so the cascade stops well short of conversion even though endoglucanase and cellobiohydrolase are both present and active. It is a product-inhibition cascade of exactly the kind part 2 chapter 3 describes, and it is the reason β-glucosidase is supplemented into commercial cocktails and the reason simultaneous saccharification and fermentation works.

3. What makes consolidated bioprocessing hard, in one sentence, with the module reference?

It requires one organism to be both a high-capacity protein secretor and an efficient fermenter of mixed sugars, and those are different physiologies — a cellulolytic organism engineered to ferment runs into its own tolerance and pathway limits, while a good fermenter engineered to secrete a multi-enzyme cocktail runs into the secretion-capacity and proteome-cost limits of part 6 chapter 59 and part 1 chapter 9.

What to open a book for. A current review of lignocellulose pretreatment for the severity comparisons, and one of cellulase cocktails and LPMOs for the enzymology — this is a field where the reviews are better than the textbooks because the LPMO work is recent. The NREL biomass compositional analysis procedures are the reference for the composition numbers, and your own feedstock's composition is the first thing to measure rather than assume.

Chapter S3 · extends week 3

Two-substrate kinetics and allostery: mechanisms, MWC and KNF

Almost every enzyme you will engineer has two substrates, and single-substrate Michaelis–Menten cannot describe it. The mechanisms are distinguishable by a pattern you can see in two minutes of plotting.

Why one substrate is not enough

Part 2 chapter 3 derived $v = V_{max}[S]/(K_M + [S])$ for $E + S \rightleftharpoons ES \rightarrow E + P$. Now count the enzymes in a pathway you would actually build: every NAD(P)H-dependent oxidoreductase has two substrates, as does every transaminase, every kinase, every ligase and every acyltransferase. For those, a single-substrate fit gives you an *apparent* K_M that depends on the concentration of the other substrate — which is why two groups measuring the same enzyme report different numbers, and why a cofactor-swap campaign (part 2 chapter 19) needs the two-substrate form to be meaningful at all.

Two mechanisms, and the plot that tells them apart

Ternary-complex mechanisms — ordered or random Bi-Bi — require both substrates to be bound at once before chemistry happens. The rate law is

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v = V[A][B] / (K_ia·K_b + K_b[A] + K_a[B] + [A][B])
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Ping-pong (substituted-enzyme) mechanisms take the first substrate, release the first product carrying a group on the enzyme, then bind the second — so the two substrates are never on the enzyme together, and the $K_{ia}K_b$ term disappears:

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v = V[A][B] / (K_b[A] + K_a[B] + [A][B])
```

That one missing term is visible. Plot 1/v against 1/[A] at several fixed [B] and read the slopes. Computed for V = 1, K_a = 0.5, K_b = 0.4, K_{ia} = 0.6:

Fixed [B]	Ternary complex		Ping-pong	
	slope	intercept	slope	intercept
0.2	1.700	3.000	0.500	3.000
0.5	0.980	1.800	0.500	1.800
1.0	0.740	1.400	0.500	1.400
4.0	0.560	1.100	0.500	1.100

Ternary complex gives intersecting lines; ping-pong gives parallel lines. The reason is in the algebra: the slope with respect to 1/[A] is $(K_a + K_{ia}K_b/[B])/V$ for the ternary mechanism and K_a/V for ping-pong, so only the first depends on [B]. Transaminases and many flavin enzymes are ping-pong, because the group is parked on the cofactor between half-reactions; most NAD(P)-dependent dehydrogenases are ordered ternary, with the cofactor binding first.

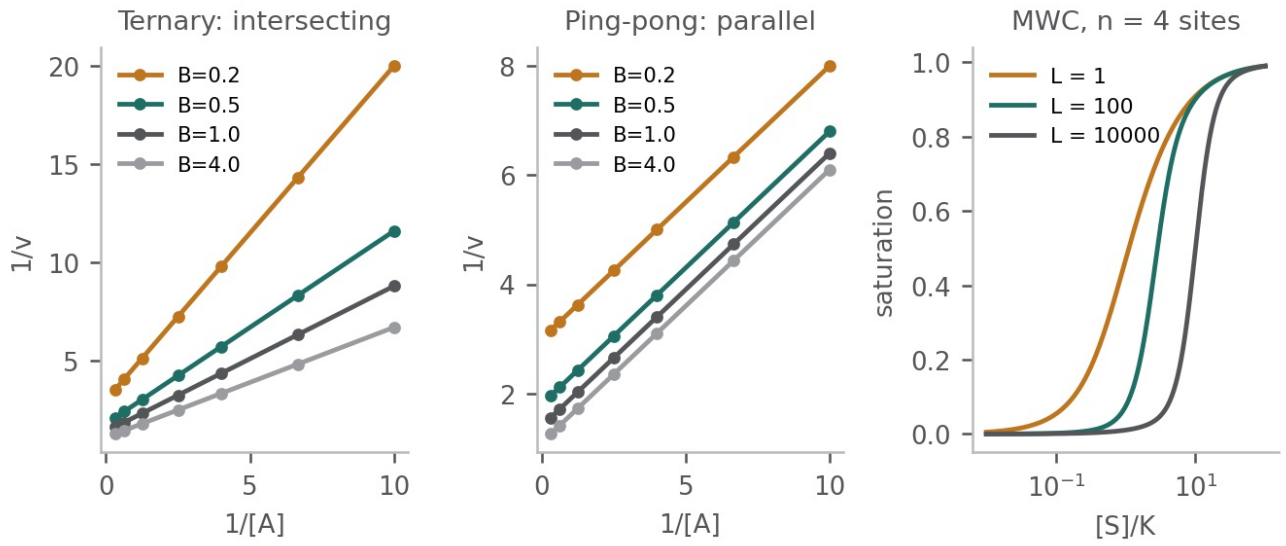


Figure S3.1 Left and centre: double-reciprocal plots for the two mechanisms at the stated parameters — intersecting against parallel. Right: the MWC saturation curve for an enzyme with four sites at three values of the allosteric constant L. Computed for this module; these are exact rate laws rather than fitted data.

Two practical consequences. The **order of a fit matters**: fitting a ping-pong enzyme with a ternary rate law will return a K_{ia} near zero and a large standard error, which is the model telling you which mechanism it is. And the **design question differs**: in an ordered mechanism where the cofactor binds first, a cofactor swap changes the first binding event and therefore the whole catalytic cycle, while in a ping-pong mechanism the two half-reactions can be engineered more nearly independently.

Where the Hill coefficient comes from

Part 2 chapter 3 used the Hill equation as an empirical description of a sigmoid and said n is a measure of cooperativity rather than a number of sites. The two models behind it say why.

MWC (Monod–Wyman–Changeux), the concerted model. The enzyme exists in two conformations, T (tense, low affinity) and R (relaxed, high affinity), and *all subunits switch together* — no hybrid states. Three parameters: n, the number of sites; L, the equilibrium constant [T]/[R] with no ligand bound; and c = K_R/K_T, the affinity ratio. Substrate binds preferentially to R, so each binding event shifts the T⇌R equilibrium toward R, which raises the affinity of the remaining sites. That indirect coupling is the whole mechanism — the sites never touch.

Computed for n = 4 sites and c = 0.01:

L (T/R with no ligand)	Half-saturation at [S]/K	Apparent Hill coefficient
1	1.11	1.12
10	1.59	1.47
100	2.85	2.04
1,000	5.34	2.58
10,000	10.00	2.78

Three things to take from that table. The apparent Hill coefficient **never reaches 4**, the number of sites — it approaches it only in a limit that real enzymes do not occupy, which is the quantitative justification for part 2 chapter 3's warning. Cooperativity requires the protein to *start* mostly in T: at L = 1 the enzyme is effectively non-cooperative. And an allosteric activator or inhibitor works by changing L — an inhibitor raises it, pushing the enzyme into T and both lowering the rate and sharpening the switch, which is why feedback inhibition and cooperativity so often appear on the same enzyme.

KNF (Koshland–Némethy–Filmer), the sequential model. Subunits change conformation one at a time, on binding, and the change alters the interaction energy with neighbours. Because the subunits are allowed to differ, KNF can produce **negative** cooperativity — a Hill coefficient below 1, where each binding event makes the next harder — which MWC structurally cannot. So the experimental signature that distinguishes them is not a better fit to a sigmoid; it is the sign of the cooperativity, with negative cooperativity ruling MWC out.

For engineering, the useful consequence is the one part 2 chapter 3 already used without derivation: a **feedback-**

resistant variant is a mutation at the allosteric site that stops the effector binding, so L no longer responds to the end product. The enzyme keeps its catalytic site and loses its regulation, which is why the usual outcome is full activity in the presence of the product and why those variants are the single most effective change in amino-acid strains.

Ternary complex	$v = V[A][B]/(K_{ia}K_b + K_b[A] + K_a[B] + [A][B])$
Ping-pong	$v = V[A][B]/(K_b[A] + K_a[B] + [A][B])$
Diagnosis	1/v against 1/[A] at several [B]: intersecting = ternary, parallel = ping-pong
MWC	all subunits switch together; parameters n, L, c; apparent Hill < n always
KNF	sequential; permits negative cooperativity, which MWC cannot
Feedback resistance	a mutation that stops the effector changing L

You measure an NADPH-dependent reductase and want to swap it to NADH. What do you measure, and in what form?

Not a single apparent K_M. Measure the full two-substrate surface — initial rates over a grid of NADPH and ketone concentrations — and fit the ternary rate law, since a dehydrogenase of this class is almost certainly ordered with the cofactor binding first. That gives V, K_a, K_b and K_{ia}, and the double-reciprocal pattern confirms the mechanism before you trust the parameters. Then repeat with NADH, and report the **specificity constants k_{cat}/K_M for each cofactor**, because the swap is a change in their ratio and nothing else is comparable between variants.

Two things the two-substrate form tells you that the single-substrate one hides. If the cofactor binds first in an ordered mechanism, its K_{ia} — the dissociation constant from the binary complex — is the quantity your pocket redesign is actually changing, and it is invisible in an apparent K_M. And because the apparent K_M for the ketone depends on the cofactor concentration, a variant that looks improved at a saturating cofactor concentration may be worse at the intracellular one — which, with part 1 chapter 4's point that the two cofactor pools sit at very different ratios, is the difference between a variant that works in a cuvette and one that works in a cell.

Three things people get wrong

- 'The K_M of the dehydrogenase is 40 µM.' For which cofactor concentration? A two-substrate enzyme has no single K_M.
- 'The Hill coefficient is 2, so there are two sites.' MWC with four sites gives apparent Hill coefficients from 1.1 to 2.8 depending on L alone.
- 'MWC and KNF are interchangeable descriptions.' Negative cooperativity is possible in KNF and impossible in MWC, which is the experiment that separates them.

Self-check — answer on paper first

1. Your double-reciprocal plots at four fixed [B] are parallel. What mechanism, and what does it predict about inhibition?

Ping-pong: the slope with respect to 1/[A] is K_a/V and contains no [B], which is what parallel lines mean. It predicts that the first product is released before the second substrate binds, so that product should be a *competitive* inhibitor with respect to the first substrate — and testing that is the confirmation, because the parallel-line pattern alone can also arise from a ternary mechanism with a very small K_{ia}. It also means the enzyme passes through a covalently modified intermediate, which is often assayable in its own right.

2. Why can an MWC enzyme with six sites never show a Hill coefficient of six?

Because the apparent Hill coefficient measures the steepness of the observed curve, and the observed curve is a weighted mixture of the T and R binding curves rather than a pure all-or-nothing transition. Only in the limit of infinite L and infinitesimal c — the enzyme starting entirely in T and R binding infinitely more tightly — does the mixture approach a step with slope equal to the number of sites; at any finite L a fraction of the enzyme is already in R before ligand arrives, and that fraction binds non-cooperatively and flattens the curve. The computed table shows the approach: n = 4 gives 2.78 at L = 10,000.

3. A feedback-resistant variant has the same k_{cat} and no inhibition by the end product. In MWC terms, what changed?

The effector can no longer bind, or can no longer couple its binding to the conformational equilibrium, so the end product stops raising L. The enzyme's catalytic site and therefore its k_{cat} are untouched, which is exactly what you want and exactly what makes these variants effective; what is lost is the regulation that protected the cell from accumulating its own end product, which is the cost part 2 chapter 3 names.

What to open a book for. Cornish-Bowden, *Fundamentals of Enzyme Kinetics*, chapters on multi-substrate mechanisms and on cooperativity — this is the book the module has cited four times and this is the chapter where you actually need it. Segel's *Enzyme Kinetics* if you want every mechanism derived. For the allosteric models, the original MWC paper (J. Mol. Biol., 1965) is short, readable and still the clearest statement of the concerted idea.

Chapter S4 · extends weeks 9, 26 and 27

Synthetic consortia: division of labour, and why it does not stay divided

Splitting a burdensome pathway across two strains is a direct answer to the proteome problem of week 9. It is also an ecology problem, and the ecology has a cheater in it.

The argument, quantified

Part 1 chapter 9 established that expressing a pathway costs proteome and that growth falls roughly in proportion to the fraction diverted. If a pathway costs a fraction φ of the proteome and you split it across two strains, each carries φ/2 and each grows faster:

Total pathway proteome cost	One strain, μ	Split across two, μ each	Gain
10%	0.540 h^{-1}	0.570	+6%
20%	0.480	0.540	+13%
30%	0.420	0.510	+21%

The gain is real and modest, and it is largest exactly where the single-strain route hurts most. Three other arguments usually matter more than the growth number. **Incompatible conditions**: a pathway needing an anaerobic step and an aerobic one, or two enzymes with pH optima three units apart, can be split so each half runs where it works. **Compartment-free compartmentalisation**: a toxic intermediate can be made in one strain and consumed in another, which is part 2 chapter 19's problem solved by ecology rather than by a membrane. And **modularity**: each strain is a smaller construct, so the design–build–test cycle per module is shorter, which by part 2 chapter 11's arithmetic is the highest-value property a project can have.

The cost is the interface. The intermediate must leave one cell and enter the other, which is part 2 chapter 14's transport problem twice over — and if it is charged, it needs an exporter and an importer, not diffusion.

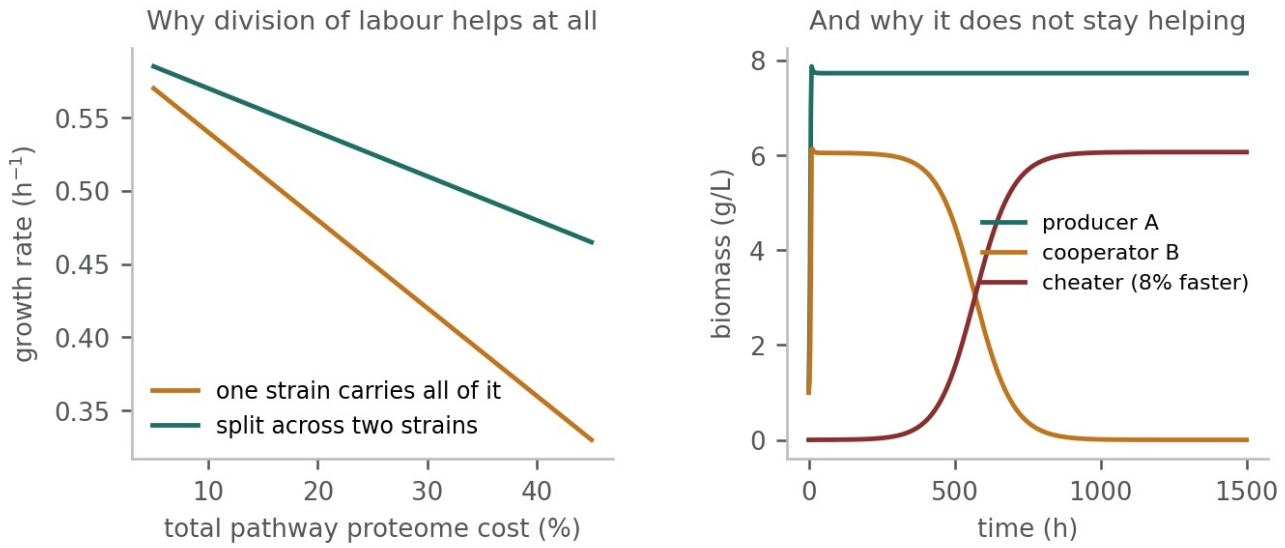


Figure S4.1 Left: growth rate against total pathway proteome cost, carried by one strain or split across two. Right: a chemostat simulation of obligate cross-feeding invaded by a cheater that stops making its own costly enzyme and grows 8% faster. Computed for this module from the model in the text; the 8% advantage and the burden slope are assumptions.

Keeping the ratio where you want it

The population ratio is not a design parameter you set — it is a steady state the ecology chooses. For obligate cross-feeding, where strain A makes an intermediate that strain B needs, a chemostat model makes this concrete:

```
dS/dt = D(S₀ − S) − r_A·A/Y_A
dI/dt = −D·I + α·r_A·A − r_B·B/Y_B          I is the shared intermediate
dA/dt = (r_A − D)·A        dB/dt = (r_B − D)·B
```

At D = 0.2 h^{-1} with the parameters in the code, the steady state is A = 7.73 and B = 6.05 g/L, a ratio of **1.28** — set by the yields and the intermediate's stoichiometric coefficient α, not by the inoculum. Changing the dilution rate barely moves it (1.25 at D = 0.05, 1.28 at 0.2) until washout approaches, where it climbs to 2.56 at D = 0.45 because the slower partner is washing out first.

Two readings. The ratio is **self-correcting** under obligate mutualism, which is the attractive property: if B falls behind, the intermediate accumulates, B grows faster and the ratio restores. And the **washout condition is set by the slower partner**, so the consortium's operating window is narrower than either strain's alone — a point to check before proposing continuous operation.

The cheater, which is the real objection

Mutualism is only stable if cooperating is not optional. Add to the same chemostat a variant of B that has lost its costly contribution — a leaky auxotroph complementation, a deleted export enzyme, a silenced pathway module — and grows 8% faster as a result:

Time	Producer A	Cooperator B	Cheater	Cheater share of the B-type population
0 h	1.00	1.000	0.0001	0%
200 h	7.73	6.036	0.017	<1%
500 h	7.73	4.533	1.525	25%
1,000 h	7.73	0.010	6.059	100%

From one part in ten thousand to complete takeover in about a thousand hours — forty days, which is inside a long continuous campaign and well inside a scale-up seed train plus production run. An 8% advantage is not large; it is the same order as the burden numbers above, which is the uncomfortable symmetry: **the thing that made division of labour worth doing is exactly what makes cheating profitable**.

What stops it, and what each costs:

- **Obligate mutual dependency.** Make each strain auxotrophic for something only the other supplies, so a cheater cannot grow at all rather than merely paying less. This is the only defence that is structural rather than quantitative, and it is why engineered consortia are usually built from reciprocal auxotrophs. It costs a growth penalty on both strains and makes the medium a design parameter.
- **Growth coupling within each strain**, as in part 11 chapter 39: if the contribution is on the path to the strain's own growth, losing it is not an advantage.
- **Spatial structure.** Biofilms and droplets keep a cooperator's product near its relatives, so the benefit is not freely shared; microfluidic droplet compartments do this deliberately. It costs the homogeneity a stirred tank provides.
- **Short campaigns and fresh inocula**, which is the honest industrial answer: run batches short enough that the cheater cannot sweep, and sequence the end of the run — part 7 chapter 32's instability argument, now with a second mechanism.

Designing one, and the vocabulary reviewers expect

The terms to use correctly: **division of labour** for splitting a function; **cross-feeding** for one strain's metabolite supporting another, **obligate** if the recipient cannot grow without it and **facultative** otherwise; **mutualism**, **commensalism** and **parasitism** for who benefits; **auxotroph complementation** for the reciprocal-dependency design; **cheater** or **defector** for the variant that takes the benefit without paying; and **coexistence** as the property to demonstrate rather than assume. The ecological point reviewers will test is whether you know that a consortium is a population with its own dynamics, not a pair of strains in the same vessel.

The minimum experiment that demonstrates one works: the population ratio over time by a strain-specific marker — differential plating or qPCR on a unique locus, not optical density, which cannot distinguish them; the titer against the best single-strain control, since a consortium that only matches it has bought nothing; and the ratio and titer after a deliberately long run or a serial-transfer experiment, which is the only test for a cheater sweep. Reporting the first two and not the third is the commonest gap in consortium papers.

Burden split	`μ(φ) → μ(φ/2) per strain; +13% growth at φ = 20%`
Cross-feeding steady state	`ratio set by yields and the stoichiometric coefficient, not the inoculum — 1.28 here`
Washout	`set by the slower partner; the window is narrower than either strain's`
Cheater sweep	`an 8% advantage goes from 10⁻⁴ to fixation in about 1,000 h`

The only structural defence	`reciprocal auxotrophy — a cheater cannot grow, rather than merely paying less`

A proposal splits a twelve-enzyme pathway across three strains to reduce burden. What are your three questions?

What crosses between them, and how? Two interfaces for three strains, each needing an intermediate to leave one cell and enter another — and if either intermediate is charged, that is two exporters and two importers to find or engineer, which is part 3 chapter 45's screening problem four times over. A proposal that does not name the transporters has not costed the interface.

What stops the cheater? With three strains there are three ways to defect and the simulation above says an 8% advantage fixes in about a thousand hours. If the answer is not reciprocal auxotrophy or growth coupling, the answer is a short batch and sequencing at harvest, which should be stated.

What is the control? The comparison is against the best single-strain strain, tuned, not against an untuned one — and against a two-strain split, because the burden gain from splitting is sublinear while the number of interfaces grows linearly, so three strains may well be worse than two. I would also ask for the ratio measurement method up front, since optical density cannot see it.

Three things people get wrong

- ‘We set the population ratio at inoculation.’ The ratio is a steady state the ecology chooses; the inoculum sets only how long it takes to get there.
- ‘The consortium is stable because it is mutualistic.’ Mutualism is stable against extinction and not against cheating, and cheating is profitable in proportion to the burden you were trying to avoid.
- ‘Splitting reduces burden, so it increases titer.’ It increases growth rate per strain; whether titer follows depends on the interface, and the control is a tuned single strain.

Self-check — answer on paper first

1. Why is reciprocal auxotrophy qualitatively better than growth coupling for consortium stability?

Because it changes the cheater's payoff from a quantitative penalty to zero growth. Under growth coupling the defector still grows, just less well, so the defence depends on the coupling being tight enough that the advantage is negative — a quantitative claim that a suppressor mutation can undo. Under reciprocal auxotrophy the cheater lacks something it cannot make and no partner will give it, so it cannot grow at all in that medium. The cost is that both strains grow more slowly than prototrophs and the medium must not contain the complemented nutrients, which constrains the feed.

2. Your two-strain consortium drifts from 1:1 to 4:1 over a fermentation. Is that a failure?

Not necessarily — the ratio is a steady state set by the yields and the intermediate stoichiometry, so a drift away from the inoculated ratio toward the model's prediction is the system working as designed. It is a failure if the drift continues past the predicted steady state, which points to a cheater sweep or to one strain losing its construct, and the test distinguishes them: plate and screen for the phenotype, or sequence the locus. The thing not to do is re-inoculate to 1:1 and call it fixed.

3. What is the minimum evidence that a consortium beats a single strain?

Titer and yield against a *tuned* single-strain control carrying the whole pathway, run in the same vessel and medium; the population ratio over time by a strain-specific marker, showing it reaches and holds its steady state; and the same two measurements after a long run or serial transfer, which is the only test of whether the advantage survives the ecology. Without the third, a consortium paper reports a result that may not exist at the end of a campaign.

What to open a book for. A recent review of synthetic microbial consortia for the design patterns and the vocabulary; the reciprocal-auxotrophy literature for the stability argument; and, if you want the dynamics properly, any mathematical-ecology text's treatment of chemostat competition — the equations are the ones in this chapter and the analysis is a century old.

Chapter S5 · extends weeks 43 and 51

Measuring the pools and the proteins: quantitative metabolomics and proteomics

Part 2's thermodynamics and part 4's enzyme constraints are only as good as the concentrations and abundances fed into them. This chapter is about where those numbers come from and how wrong they are.

Quenching: the measurement that is over before you start

Part 2 chapter 13 argued that metabolite pools are small and turn over fast, which is what justifies the pseudo-steady state. Read it backwards and it is a measurement problem: a pool's turnover time is its size divided by the flux through it, and anything slower than that changes the pool while you are sampling it.

Pool	Flux	Turnover time	Fraction turned over in a 1 s quench
0.1 mM	1.0 mM/s	0.10 s	essentially all of it
0.5 mM	2.0 mM/s	0.25 s	essentially all of it
2.0 mM	1.0 mM/s	2.0 s	about 40%
10 mM	0.5 mM/s	20 s	about 5%

So for central-metabolism intermediates, **sampling must be faster than a second**, which is why the standard method is a quench directly into cold methanol at −40 °C and why rapid-sampling devices exist on fermenters. The two artefacts to know, because they pull in opposite directions: cold methanol **leaks** small polar metabolites out of the cells, so a separated cell pellet under-reports them; and whole-broth extraction avoids the leakage and then measures intracellular and extracellular pools together, which for a secreted product is meaningless. There is no method without an artefact — only a method whose artefact you have quantified, which is the same conclusion as part 4 chapter 50's extraction discussion and is worth stating in a methods section rather than hiding.

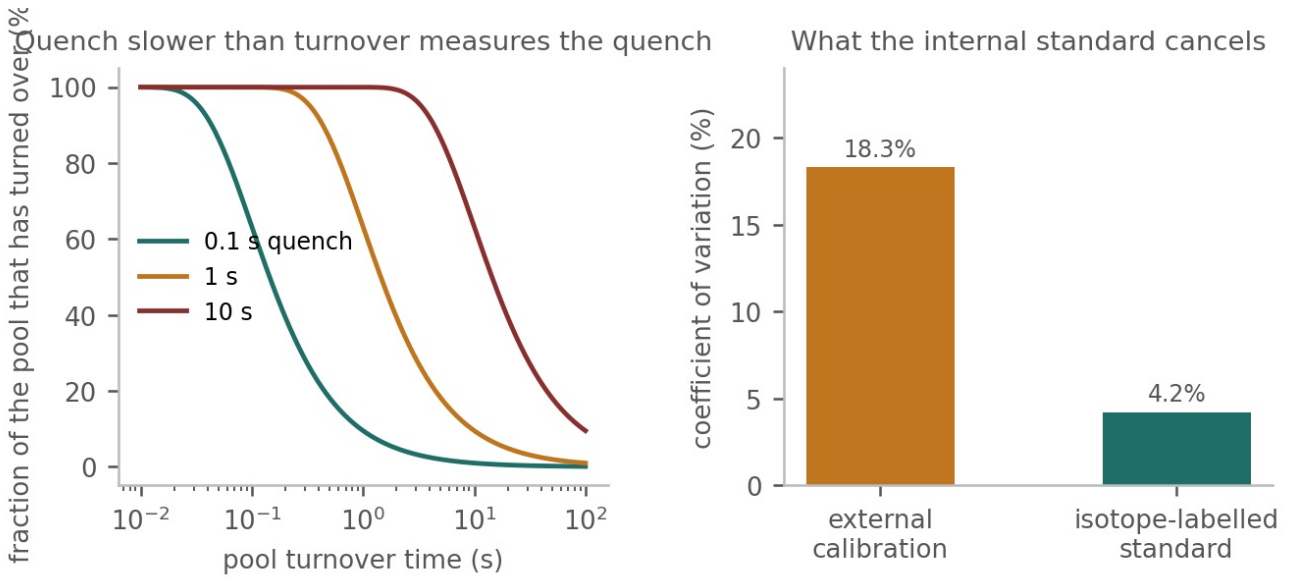


Figure S5.1 Left: the fraction of a pool that turns over during a quench of 0.1, 1 and 10 seconds, against the pool's turnover time. Right: the coefficient of variation of a quantification with external calibration alone and with an isotope-labelled internal standard, from the error model in the text. Computed for this module.

What the internal standard actually buys

Part 4 chapter 50 said an isotope-labelled internal standard is mandatory for mass-spectrometric quantification. Here is the arithmetic. Model four independent error sources — extraction recovery at 8%, ionisation suppression at 15%, injection volume at 5% and detector response at 3% — and compare a quantification against an external calibration curve with one against a co-eluting labelled standard that experiences extraction, suppression and injection identically:

```
external calibration only  : CV 18.3%
isotope-labelled standard  : CV  4.2%
```

A factor of four, and it comes from cancellation rather than from precision: the labelled standard shares every error that acts on both species, so the ratio retains only what differs between them. That is also the limit of the method — it cannot cancel an error the standard does not share, such as incomplete quenching, degradation before the standard is added, or a matrix effect that differs across the chromatographic peak. **Add the standard as early in the workflow as possible**, ideally into the quenching solvent, because every step before it is unprotected.

Targeted against untargeted, and identification confidence

	Targeted (MRM on a triple quadrupole)	Untargeted (high-resolution MS)
What it measures	A defined list, typically tens to a few hundred compounds	Thousands of features, most unidentified
Quantification	Absolute, with labelled standards per analyte	Relative, and semi-quantitative at best
Use	Feeding a model: thermodynamics, kinetics, flux fitting	Hypothesis generation; finding the unexpected accumulation
Main risk	You only see what you looked for	Identification, and the statistics below

Identification is the part that engineers under-rate. A feature is a mass and a retention time, not a compound, and the community's confidence levels run from a mass match alone — the weakest, and the most commonly over-claimed — through a spectral-library match, to confirmation against an authentic standard run on the same instrument, which is the only level at which "we detected X" is a measurement. **A pathway intermediate reported from an untargeted run without a standard should be treated as a candidate**, and confirming it is a day's work with a purchased standard.

The statistics of many metabolites and few samples

Simulate 200 metabolites with four replicates, where 20 have truly changed by 0.8 in $\log_2$ and the measurement CV is 35%:

Standard error of $\log_2$ fold change	Raw $p < 0.05$	False among them	BH $q < 0.05$	False	True changes found
0.36	18 calls	9	3 calls	0	3 of 20

Half the raw calls are false, and correcting properly leaves three of the twenty real changes — this is part 2 chapter 17 and part 9 chapter 33 in a new dataset, and the conclusion is the same: **four replicates of 200 metabolites is a survey, not a test**. Two further habits specific to metabolomics: normalise to a biomass measure rather than to a volume, since the pool concentration is per cell and dry weight is the denominator your balances need; and treat the data as compositional if the normalisation is to a total, because a large change in one abundant metabolite moves every other one, which is part 9 chapter 33's composition argument applied to pools rather than transcripts.

Proteomics, and the uncertainty it feeds into GECKO

Part 4 chapter 51 used $v \le k_{cat} \cdot [E]$ as a constraint and part 11 chapter 39 built a proteome-budget model on it. Both need absolute protein abundances, so here is how they are obtained.

Bottom-up is the default: digest the proteome with trypsin, separate the peptides, and infer proteins from the peptides identified — which carries the inference problem that a peptide shared between homologues cannot be assigned. The quantification options, in increasing order of effort: **label-free**, comparing intensities between runs, cheapest and least precise; **isobaric labelling** (TMT, iTRAQ), multiplexing samples into one run with good precision and a known ratio-compression bias; **data-independent acquisition** (DIA/SWATH), fragmenting everything in windows and quantifying from a library, which gives better reproducibility across many samples; and **absolute quantification** — AQUA peptides or QconCAT constructs, which spike known amounts of labelled peptide and are the only way to get molecules per cell rather than a ratio.

That last distinction is the one that matters for a model. An enzyme-constrained model needs **absolute** abundance, and most published proteomics is relative, so the usual practice is to anchor a relative dataset to a few absolutely quantified proteins or to a total protein mass — and the resulting uncertainty propagates straight into every flux bound. Taken together with part 4 chapter 51's warning about k_{cat} values:

```
v ≤ k_cat·[E]:   a k_cat wrong by 10× makes the bound wrong by 10×
                 an abundance with a 25% CV gives a ±25% band on every bound
```

Since the two multiply and the k_{cat} error dominates, an enzyme-constrained model's bounds are good to a factor of a few at best. That is not a reason to avoid them — a bound good to a factor of three still rejects a design needing 28% of the proteome — but it is a reason to report the model's conclusions as what they are: statements about what is impossible, with an uncertainty set by the worst k_{cat} in the pathway.

Quench speed	must beat the pool's turnover time, pool/flux — under a second for central metabolism
Internal standard	CV 18.3% → 4.2%; cancels only errors it shares, so add it into the quench
Identification	mass match < library match < authentic standard; only the last is a measurement
200 metabolites, n = 4	9 of 18 raw calls false; BH leaves 3 of 20 real changes found
Proteomics for models	absolute quantification (AQUA/QconCAT), not relative, or the bound has no units
Bound uncertainty	k_cat error × abundance error; the first dominates and is often 10×

You need intracellular concentrations for the max-min driving force calculation of part 2 chapter 14. What do you actually do?

Targeted, with labelled standards, on the dozen or so intermediates the pathway contains — not an untargeted run, because what you need is absolute concentrations for a specific list and untargeted gives relative intensities for thousands of features. The sequence: rapid sampling into cold methanol at −40 °C with the labelled standards already in the quenching solvent; a decision about whole-broth against separated cells, made and stated, with the extracellular concentration measured separately so the intracellular can be inferred if you chose whole broth; extraction, with recovery measured per compound by a second spike; and LC-MS/MS in MRM mode against a calibration in matrix.

Then the two conversions people forget. Concentrations come out per gram of dry weight or per millilitre of extract, and $\Delta G'$ needs **molar concentrations in the cytoplasm**, so you need the cell volume per gram of dry weight from sheet 2 of the lookup appendix — an assumption that carries directly into every $\Delta G'$. And cofactor ratios, not absolute cofactor concentrations, are what enter Q for a redox reaction, so NAD^+/NADH and $NADP^+$/NADPH must be measured in the same sample, which is harder than it sounds because the reduced forms degrade quickly at low pH and the oxidised forms at high pH — the usual answer being two extractions in parallel.

Three things people get wrong

- ‘We quenched quickly.’ A 0.1 mM pool at 1 mM/s turns over in 0.1 s; anything slower measured the quench, not the cell.
- ‘The feature at m/z 259.02 is the intermediate.’ A mass is a candidate. Only an authentic standard on the same instrument makes it a measurement.
- ‘Our proteomics gives the enzyme abundances for the model.’ Relative abundance has no units; a flux bound needs absolute, and the k_{cat} is still the dominant error.

Self-check — answer on paper first

1. Why does an isotope-labelled internal standard reduce the CV fourfold, and what does it fail to cancel?

Because it is chemically identical to the analyte and therefore experiences the same extraction recovery, the same ionisation suppression and the same injection variability, so taking the ratio removes every error the two share and leaves only what differs — essentially the detector's own noise on two nearby masses. It fails to cancel anything that happens before it is added, which is why it goes into the quenching solvent, and anything that affects the two species differently: incomplete quenching, degradation of the analyte by a route the standard does not take, and isotope effects on chromatography, which can separate the labelled and unlabelled peaks enough that they see different matrix.

2. A collaborator reports 40 significantly changed metabolites from 200 measured, n = 3, raw p < 0.05. What do you say?

That about ten of them are expected by chance from the nulls alone — 5% of roughly 180 unchanged metabolites — and that the list has had no multiple-testing correction. Ask for Benjamini–Hochberg-adjusted values and expect the list to shrink to a handful, as the simulation in this chapter did from 18 to 3. Then ask two metabolomics-specific questions: what the normalisation was, since normalising to a total makes the data compositional and a large shift in one abundant metabolite moves all the others; and whether the identifications are standard-confirmed, since an unidentified feature cannot be interpreted however significant it is.

3. What is the uncertainty on a flux bound in an enzyme-constrained model, and how should it be reported?

The product of the k_{cat} uncertainty and the abundance uncertainty, dominated by the first: literature k_{cat} values are frequently wrong by an order of magnitude for the organism and conditions you care about, while absolute abundance might carry 25%. So a bound is good to a factor of a few. Report it as a bound with its assumed k_{cat} and source named, state the conclusion as what the model excludes rather than predicts, and show the sensitivity — if the conclusion survives a tenfold change in the worst k_{cat}, say so, because that is the statement a reader can rely on.

What to open a book for. A review of microbial metabolomics sample preparation — the quenching and extraction comparison is the content, and the reviews are better than any textbook because the artefact comparisons are experimental. For proteomics, a bottom-up methods review covering label-free, TMT and DIA, and the AQUA/QconCAT papers for absolute quantification. The metabolomics community's identification-confidence levels are worth reading in the original short note that defined them.

Chapter S6 · extends weeks 16, 27 and 44

Control theory for pathways: stability, bistability, integral feedback, optimal feeding

Part 2 integrated models and part 7 raised loop stability without analysing it. The analysis is two lines of linear algebra, and it answers questions the simulation cannot: whether a design oscillates, whether it switches, and whether it can reject a disturbance perfectly.

Stability from the Jacobian

Integrating a model tells you what it does from one initial condition. Linearising it tells you what it does near a steady state, for every initial condition. Take the three-enzyme pathway of part 2 chapter 16, compute the Jacobian J = ∂f/∂c numerically at the steady state, and read its eigenvalues: **negative real parts mean stable, a positive real part means unstable, and a non-zero imaginary part means the approach oscillates**. The real part's magnitude is the decay rate, so the slowest eigenvalue sets the settling time — which is the formal version of part 8 chapter 26's observation that a loop is as fast as its actuator.

Feedback cooperativity n	Eigenvalue real parts (h⁻¹)	Oscillatory?	Reading
1	−0.144, −0.369, −0.920	no	Overdamped — monotone return
2	−0.268, −0.268, −1.009	yes	Damped oscillation; fastest settling of the set
4	−0.257, −0.257, −1.176	yes	Still stable
8	−0.210, −0.210, −1.407	yes	Less damped
16	−0.123, −0.123, −1.677	yes	Approaching the imaginary axis

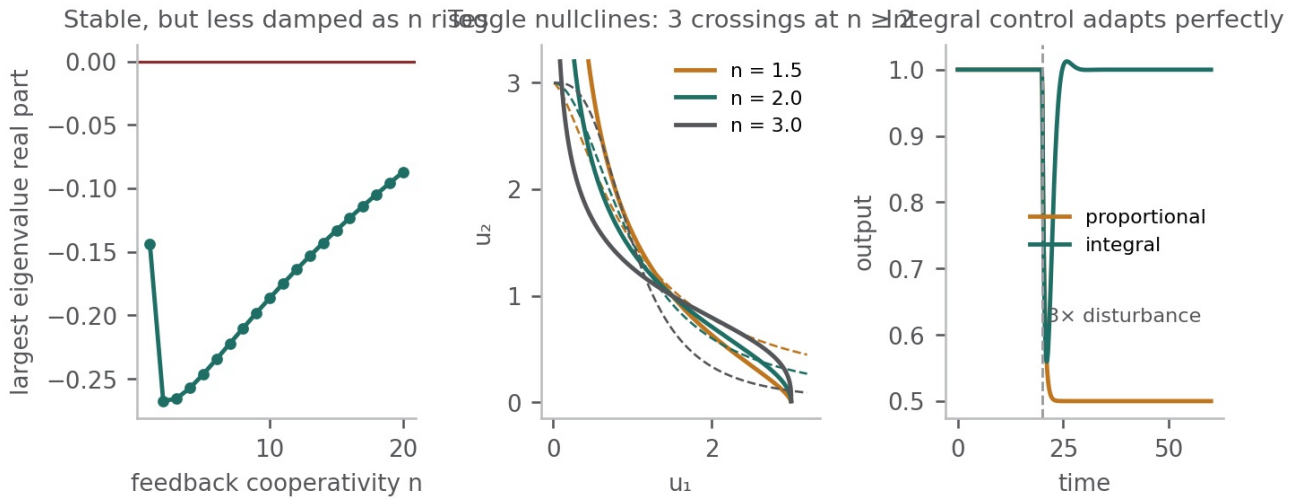


Figure S6.1 Left: the largest eigenvalue real part against feedback cooperativity for the part 2 chapter 16 pathway — stable throughout, but increasingly lightly damped. Centre: toggle-switch nullclines, which cross once at n = 1.5 and three times at n ≥ 2. Right: proportional against integral feedback under a threefold disturbance. Computed for this module.

The pattern is the one control engineers expect and biologists usually discover empirically: **sharper feedback makes a loop faster to respond and less damped**, and past some gain it would oscillate outright. This model stays stable because its feedback acts instantaneously; real circuits add the delay of transcription, translation and dilution, and delay is what converts light damping into sustained oscillation. That is the formal reason part 7 chapter 24 insisted a dynamic controller's gain and delay be considered before building it — and the measurement that tests it is the disturbance response, not the titer.

Bistability, and the condition for a switch

The toggle switch — two repressors, each repressing the other — is the canonical bistable circuit:

```
du₁/dt = α₁/(1 + u₂ⁿ) − u₁        du₂/dt = α₂/(1 + u₁ᵐ) − u₂
```

Its steady states are where the nullclines cross, and the number of crossings is the question. With α = 3 for both:

Cooperativity n = m	Steady states found	Values
1.0	1	(1.303, 1.303) — a single symmetric state
1.5	1	(1.251, 1.251)
2.0	3	(0.382, 2.618), (1.213, 1.213) unstable, (2.618, 0.382)
3.0	3	(0.108, 2.996), (1.164, 1.164) unstable, (2.996, 0.108)

Bistability appears between n = 1.5 and n = 2, and that threshold is the design rule: a switch requires cooperativity, because without it the nullclines are too shallow to cross three times however strong the repression. The middle state is a saddle — unstable, and the boundary between the two basins of attraction — so a perturbation that pushes the system past it flips the switch permanently, which is what memory means in a circuit. The transition as n crosses the threshold is a **pitchfork bifurcation**, and the vocabulary matters because "the circuit becomes bistable above a cooperativity of about 2" is a statement a reviewer can check, while "the circuit shows switching behaviour" is not.

Oscillators follow the same analysis with a different answer: a repressilator or a delayed negative-feedback loop loses stability through a **Hopf bifurcation**, where a complex eigenvalue pair crosses into the right half plane and a limit cycle appears. Glycolytic oscillations in yeast are the natural example, and the engineering relevance is that a dynamic metabolic controller with too much gain and a generation-time delay sits uncomfortably close to that boundary.

Integral feedback and perfect adaptation

Here is the result worth taking from control theory into circuit design. A proportional controller — one whose action is proportional to the error — leaves a steady-state offset: the error is what drives the action, so it cannot be zero. An **integral** controller acts on the accumulated error, so the only steady state it admits is zero error, whatever the disturbance. Simulating both under a threefold step disturbance:

Controller	Output before	Output after a 3× disturbance	Steady-state error
Proportional	1.000	0.500	**50%**
Integral	1.000	1.000	**0%**

This is **robust perfect adaptation**, and the robustness is structural: it does not depend on the parameter values, only on the integral being there. Biology implements it with an **antithetic** motif — two species that annihilate each other, so that their difference integrates the error — and bacterial chemotaxis is the textbook natural example, where the methylation state integrates the receptor occupancy and the cell returns exactly to its baseline tumbling frequency at any background attractant concentration.

For a pathway, the consequence is the design target part 7 chapter 24 named without the theory: if you want an intermediate held at a set point regardless of how the feed or the growth rate drifts, a proportional sensor-actuator loop will not do it — it will hold the intermediate near a value that moves with the disturbance — and an integral implementation will. It costs two more components and a slower response, since integral action trades speed for accuracy, and it can destabilise a loop that already has delay, which is the standard trade in any controller.

Optimal feeding, briefly

Part 11 chapter 44 gave feeding strategies — constant, exponential, DO-stat — as a menu. Optimal control asks the harder question: given the dynamics and a constraint on the feed, what is the feed profile that maximises product at a fixed time? Pontryagin's maximum principle answers it by introducing adjoint variables and showing that the optimum is **bang–singular–bang**: feed at the maximum rate, then on a singular arc where an interior condition holds, then at the maximum or zero again. The singular arc is usually the interesting part, and for a fed-batch it typically corresponds to holding the specific growth rate — or the specific uptake rate — at a constant value, which is precisely the exponential feed of part 11 chapter 44. **So the conventional strategy is the optimal one under those assumptions**, which is worth knowing both because it justifies the convention and because it tells you when the convention fails: when the objective is not simply final product, when the constraint is oxygen rather than feed rate, or when the product inhibits, the singular arc is no longer a constant specific rate.

Model predictive control is the implementable version: solve the optimisation over a horizon, apply the first move, measure, and re-solve. It needs a model good enough to predict a few hours ahead, which for a fermentation usually means a simple kinetic model updated from online measurements rather than a genome-scale one — and it is where a chemical engineer's training transfers most directly into a bioprocess group.

Stability	`eigenvalues of J = ∂f/∂c: negative real parts stable, imaginary parts oscillatory`

Settling time	`set by the slowest eigenvalue; sharper feedback is faster and less damped`
Bistability	`three nullcline crossings; appears between n = 1.5 and 2 for the toggle switch`
Bifurcations	`pitchfork into bistability; Hopf into a limit cycle`
Integral feedback	`zero steady-state error, structurally; antithetic motifs implement it`
Optimal feed	`bang–singular–bang; the singular arc is usually a constant specific rate`

Your dynamic controller holds the intermediate at 0.8 mM in a batch and at 1.4 mM in a fed-batch. Diagnose it in control terms.

It is a proportional controller meeting a different disturbance. The loop's action is proportional to the deviation from its implicit set point, so the steady-state error depends on the load — the rate at which the intermediate is being produced — and a fed-batch loads it differently from a batch. That is not a tuning failure; it is the structural property the simulation above shows as a 50% offset under a threefold disturbance, and no amount of gain removes it, although more gain shrinks it and brings the loop closer to oscillation.

The fix, if the set point genuinely matters, is integral action: an antithetic pair whose difference integrates the error, which gives zero steady-state error regardless of the load, at the cost of two components and a slower response. Before building it, two checks. Measure the **disturbance response** — step the feed and record the intermediate's trajectory and settling time — because that is what identifies the loop's dynamics and it is the measurement part 7 chapter 24 says almost no paper reports. And estimate the **delay**: if the actuator acts by dilution at a generation time, adding integral action to a delayed loop is the standard route to a sustained oscillation, and the right answer may be a degradation tag first and integral control second.

Three things people get wrong

- ‘The simulation is stable, so the design is stable.’ One trajectory from one initial condition; the eigenvalues tell you about all of them, and the imaginary parts tell you whether it rings.
- ‘Stronger feedback makes the loop more stable.’ It makes it faster and less damped; with delay, past some gain it oscillates.
- ‘We tuned the controller and the offset remains.’ A proportional loop has a load-dependent offset by construction. Only integral action removes it.

Self-check — answer on paper first

1. Your Jacobian has eigenvalues −0.05 ± 0.4i and −2.0. Describe the behaviour and the risk.

Stable but very lightly damped: the complex pair has a small negative real part, so a perturbation decays with a time constant of 1/0.05 = 20 time units while oscillating with angular frequency 0.4, meaning about a dozen visible cycles before it settles. The third eigenvalue is fast and irrelevant. The risk is that any additional delay, or any increase in feedback gain, pushes that pair across the imaginary axis into a Hopf bifurcation and a sustained oscillation — so the design has very little margin, and in a real circuit, where transcription, translation and dilution all add delay the model may not contain, it may already be oscillating.

2. Why does a switch need cooperativity, in terms of nullclines?

Because bistability requires the two nullclines to cross three times, and the number of crossings depends on how steeply each variable's nullcline falls. With n = 1 the repression curve is a shallow hyperbola and the nullclines can only cross once, whatever the promoter strength α. Cooperativity steepens them into a sigmoid, so each nullcline has a flat–steep–flat shape and the two can intersect at a low-u_1 state, an unstable middle state and a high-u_1 state. The computed threshold here is between n = 1.5 and n = 2, which is why toggle switches are built from dimerising repressors.

3. If the exponential feed is already optimal, why learn optimal control?

Because it is optimal only under the assumptions that make the singular arc a constant specific rate: a single feed constraint, a product objective at a fixed time, no product inhibition and no binding oxygen limit. Change any of those — add product inhibition, make oxygen transfer the constraint rather than the feed, or value the productivity rather than the final titer — and the optimal profile is no longer exponential, and only the optimisation tells you what it is. Knowing the framework also tells you what to measure to implement it: model predictive control needs a model that predicts a few hours ahead, which is a different and much more achievable thing than a model that predicts a flux map.

What to open a book for. Strogatz, *Nonlinear Dynamics and Chaos*, for stability, nullclines and bifurcations — it is the clearest treatment there is and chapters 2 to 8 cover everything in this chapter. Del Vecchio & Murray, *Biomolecular Feedback Systems*, which is free online and written for exactly this audience. Briat, Gupta & Khammash on antithetic integral feedback for the perfect-adaptation result. And your own process-control notes for Pontryagin and MPC, which is the one

section here you may already have.

Closing notes to part 13

The six chapters in one page

S1 Downstream. Recovery is multiplicative: 90% per step over eight steps is 43% overall, so deleting a step beats improving one. A 1 µm cell settles at 0.2 mm/h, which is why a centrifuge needs a sigma factor of 5,100 m² for 1 m³/h. And the capture cost per kilogram is resin price ÷ (DBC × cycle life), in which titer does not appear — titer moves the volume-driven steps only.

S2 Deconstruction. A 40/25 composition gives 0.729 kg of sugar per kg of dry biomass and a 0.373 kg ethanol ceiling. Severity collapses temperature and time into log R_0, and because release and degradation rise together the optimum is interior. Enzyme cost spans $0.13 to $13 per kg of ethanol across a plausible price range, which is why cellulosic processes live or die on it.

S3 Enzymology. Ternary-complex mechanisms give intersecting double-reciprocal lines and ping-pong gives parallel ones — the slopes computed exactly. MWC with four sites gives apparent Hill coefficients from 1.12 to 2.78 depending on L alone, which is why a Hill coefficient is not a site count, and a feedback-resistant variant is a mutation that stops the effector changing L.

S4 Consortia. Splitting a 20% pathway burden across two strains buys 13% growth. The population ratio is a steady state set by yields and stoichiometry, not by the inoculum. And a cheater with an 8% advantage goes from one part in ten thousand to fixation in about a thousand hours, which is why reciprocal auxotrophy is the only structural defence.

S5 Measurement. A 0.1 mM pool at 1 mM/s turns over in 0.1 s, so sampling must beat a second. An isotope-labelled standard takes the CV from 18.3% to 4.2% by cancellation, and cancels nothing that happens before it is added. Two hundred metabolites at n = 4 gives nine false calls out of eighteen raw ones and finds three of twenty real changes. And an enzyme-constrained bound is good to a factor of a few, dominated by k_{cat}.

S6 Control. Eigenvalues of the Jacobian give stability, oscillation and settling time; sharper feedback is faster and less damped. The toggle switch becomes bistable between n = 1.5 and 2. Integral feedback gives zero steady-state error where a proportional loop gives 50% under the same disturbance. And the exponential feed is the optimal singular arc — under assumptions worth knowing, because they fail often.

On the numbers in this part

Everything was computed with the parameters printed beside it. Four results rest on assumptions I chose and would want replaced: the illustrative release and degradation curves in figure S2.1, whose shapes set where the severity optimum falls; the 8% cheater advantage and the burden slope in S4; the four-component error model in S5's internal-standard comparison; and the biomass composition in S2, which is feedstock-specific and is the first thing to measure. Three numbers are deliberately ranges rather than values — resin price, cellulase price per kilogram of protein, and enzyme loading — because the sources disagree by an order of magnitude and a single figure from me would mislead.

Nothing here was verified against the literature, for the same reason as the rest of the module: no network access. The mechanisms, models and formulas are standard and derivable, and the figures I quote from them are my own arithmetic; the *parameter values* are the part to check, and sheet 10 of the lookup appendix is where they belong.

What is still not here

Four items from the audit I left out, with the reason. **Genetic code expansion** and **quantitative genetics** are real gaps and are reading rather than calculation — a chapter of mine would be a literature summary, which is the thing this module is least good at. **Mass-spectrometry instrumentation** is a half-day of reading that would make chapter S5 and weeks 43 and 51 read differently, and I would get it from an analytical-chemistry text rather than from me. And the **Tier 3 list** — synthetic CO_2-fixation cycles, droplet microfluidics, immobilised enzymes, single-cell omics, biophysical characterisation, intellectual property — is an awareness exercise: know the terms, know one example each, and know which would matter for your capstone. That last one, IP and freedom to operate, is the one I would actually read about before an interview with an industrial group, because it is the question a company asks that a university does not.